Crafting a Method for Counterfoil Design Research

What do you imagine when you think of a *convivial* world? For the philosopher Ivan Illich, conviviality meant "individual freedom realized in personal interdependence" and "the opposite of industrial productivity". It meant a world that centered the values of justice and personal freedom, and allowed "all people to define the images of their own future". In his book *Tools for Conviviality* [1], Illich identified a series of industrial threats to a convivial life, including economic overgrowth resulting in the concentration of enormous power in the hands of very few, the degradation of the biological environment, and forced obsolescence leading to over-consumption. Over the past fifty years, these issues have only intensified the multiple environmental, political, and social crises we are now facing [2]. The field of Computing bears a significant responsibility here, having enabled some of the most damaging changes to society and nature with the recent push towards automation and Artificial Intelligence (AI) [3]. On its current trajectory, Computing cannot pay back its debts to society [4]. As designers and Human-Computer Interaction (HCI) researchers, we cannot continue to take an uncritical approach to the tools and systems we design.

Illich's response to these crises took the form of Counterfoil research, named for how it ran counter to the dominant forms of research in service of industry [5]. Its goal was to identify when and how systems became inimical to conviviality, and then "to devise tools and tool systems that optimize the balance of life, thereby maximizing liberty for all" [1]. Today, proponents of the post-growth movement are increasingly calling for similar work that runs counter to the growth imperative [2, 6]. In this vein, I believe we need to develop methods for Counterfoil Research in the context of HCI and Design, and demonstrate its impact in enabling people to be more intentional about their technology use and the lives they create everyday.

From Friend to Foil

Design can be viewed as "the interaction between understanding and creation" [7], and my research journey has been one where a more critical understanding has over time caught up with ideas of creation. My dissertation work focuses on supporting spatial collaboration across distance by maximizing what we can achieve using everyday devices instead of head-worn displays. Headsets in their current form are expensive, far from widespread use, and contribute to exclusionary socio-technical imaginaries of what spatial computing should be [8]. In response, I have demonstrated how mobile phones can support rich and dynamic forms of spatial collaboration in a manner comparable to headset-based interaction [9]. I have conducted in-depth studies to understand how people work together using the medium of mobile Augmented Reality (AR), highlighting the ways in which mobile devices can support more embodied interaction [10]. I have designed systems for "everyday spatial computing" that enable people to use phones, tablets, and laptops to work across screen-based and spatial computing modes, making the most of whatever devices they have access to in order to meet the needs of interaction in the moment [11]. These projects have reinforced my belief in the value of using design to question dominant narratives of development.

The collective attention of HCI research and development has recently shifted away from spatial computing and towards AI. During an internship at Microsoft Research, I worked on a project that explored the potential of Generative AI to support group work across multiple meetings [12]. I began by engaging with literature from allied fields of organizational science, routine dynamics, and cognitive science to gain a better understanding of the importance of *Temporal Work* [13].

I conducted a qualitative analysis of temporal conversations in multiple series of real meetings, and created a design framework for interfaces seeking to support temporal work. I developed a prototyping tool to interrogate the ability of AI models to process temporal information based on meeting recordings, and used these insights together with the design framework to create new interface concepts. A key contribution of this project was to demonstrate the value of an approach that is grounded, not in the emergent capabilities of ever-changing models, but in an understanding of how real work actually takes place. In both of these threads across spatial computing and AI, I started out as a designer hopeful about using new technologies to help people work together. However, in both cases, I found myself questioning this technology-first mindset and seeking alternate approaches to design that serve as foils to the status quo.

The Components of a Counterfoil

Given the power wielded by those who create the tools that comprise most sociotechnical systems today, I believe a Counterfoil approach must actively and critically engage in alternative tool and system design. This echoes Phil Agre's call for a *Critical Technical Practice* [14] focusing on both the craft of design and the reflexive work of critique. While the trajectory of computing—of endless growth and increasingly undemocratic technologies—might seem inevitable when looking back and ahead, Critical Theory of Technology shows how at every point, development is contingent and shaped by complex social forces, with "multiple workable solutions to any given problem" [15]. I have found it useful to frame how Counterfoil research can help find such solutions in terms of the changes we can exert on this trajectory (Figure 1), by using critical design to give us:

1. Reasons to Think about alternative approaches. This can be through the creation of new tools [9] or methods of design [12] that question the status quo, and by learning from communities that embody convivial practices, such as makerspaces or worker-led co-ops [4].

2. Reasons to Turn, slowly but resolutely, away from the dominant trajectory. This can be done by connecting research and system design to lived experience [10], and developing strategies to reclaim our collective agency in the everyday.

3. Reasons to Try and keep working towards a convivial society. For this we need visions to strive towards that are grounded in insight from critical friends of computing [4], and in the development of alternative tools [11] that we can start using in the immediate future.

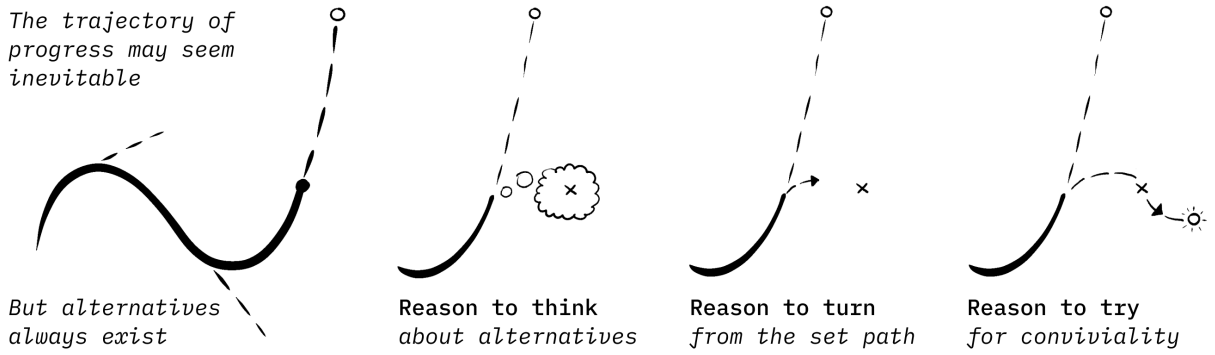


Figure 1: A graphical representation of the proposed Counterfoil approach

Why Craft a Method?

Let us consider a possible example of a Counterfoil approach applied to the state of AI tools and their integration into a range of systems and public services. Generative AI is hailed as an amplifier of human productivity and a solution to infrastructural issues around the world. A great deal of HCI research has enthusiastically followed suit [16], although some researchers continue to call for a more ethical and responsible approach to AI adoption. Applying a critical lens to these ideas reveals, once again, the imperative for incessant growth that makes the conditions of work and life increasingly precarious for the people subject to AI's use [17, 18]. Returning to *Tools for Conviviality*, Illich warns against the creation of *Radical Monopolies* where "a major tool rules out natural competence … imposes compulsory consumption and thereby restricts personal autonomy". When considering the most worrying implementations of AI today, we may well be facing the beginnings of a radical monopoly on human interaction and thought. The increased digitization, processing, and surveillance of our everyday lives lays the foundation, according to Sacasas [19], for "the enclosure of the human psyche", drawing parallels to the enclosure of the commons across Europe which heralded the rise of capitalism [2]. There is a growing awareness of the catastrophic environmental impacts of AI infrastructure that shows no intention of slowing down. Post-growth societies will be structured around digital and physical commons and respect for planetary limits to growth, goals that are fundamentally at odds with AI as currently designed. There still is latitude for a different trajectory [20] and we need to make the most of this flexibility.

To address this, we need a bottom-up approach that tackles one context of precarity at a time. We will need to identify and work with specific communities affected by the indiscriminate application of AI and collectively design more convivial tools that can meet community needs. One concrete context I am interested in is the design of Counterfoil infrastructures for the storage and sharing of media from our everyday lives. We can learn from community-led movements that support people as they switch to local or shared forms of data storage, and design new tools and workflows that reduce the barrier to moving away from platforms controlled and exploited by technology companies. Such a shift is crucial to counter the imaginaries of total information capture and control for growth at the expense of our selves and the environment. However, this approach would look very different in a different context, with different issues and community needs. It would need to be crafted to adapt based on the available levers for change, and the growth imperatives we seek to address. Counterfoils can be seen as implementations of Just Sustainability Design [4] that are biased towards the design and co-creation of tools to counter those that are being forced upon us. This approach is an extension of conversations about post-growth HCI [21], and is about creating methods that are "not repeatable but replicable … not universal but contingent … not technical but sociotechnical; not complete but proudly and savvily incomplete" [4].

In a remarkably prescient article on the struggle between authoritarian and democratic technology [22], Lewis Mumford states that "The best life possible … is one that calls for an ever greater degree of self-direction, self-expression, and self-realization. […] Life itself in its fullness and wholeness cannot be delegated". A convivial society is what would make such a life possible, and I believe that crafting a Counterfoil approach to support this is an important place for us to start.

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